

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2025

Course: B. Tech. Branch: Mechanical Engineering/Mech. Engg. (Sandwich) Semester: V

Subject Code & Name: BTME501_Y19; Heat Transfer

Max Marks: 60

Date: 21-07-2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks				
Q. 1	Objective type questions. (Compulsory Question)		12				
1	Conduction heat transfer in a solid is directly proportional to the temperature difference, area thermal conductivity and inversely proportional to thickness of solid, is the statement of		1				
	<table border="1"> <tr> <td>a. Newton's law of cooling</td> <td>b. Fourier's law</td> <td>c. Stefan Boltzmann law</td> <td>d. None</td> </tr> </table>	a. Newton's law of cooling	b. Fourier's law	c. Stefan Boltzmann law	d. None		
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2	The temperature gradient in the plane wall for heat conduction is		1				
	<table border="1"> <tr> <td>a. Linear</td> <td>b. Logarithmic</td> <td>c. Parabolic</td> <td>d. Hyperbolic</td> </tr> </table>	a. Linear	b. Logarithmic	c. Parabolic	d. Hyperbolic		
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3	Thermal conduction resistance in case of plane wall is given as		1				
	<table border="1"> <tr> <td>a. L/kA</td> <td>b. $1/hA$</td> <td>c. $1/UA$</td> <td>d. None</td> </tr> </table>	a. L/kA	b. $1/hA$	c. $1/UA$	d. None		
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4	Combined thermal resistance comprising conduction, convection and radiative resistances is represented by		1				
	<table border="1"> <tr> <td>a. Heat transfer coefficient</td> <td>b. Thermal Conductivity</td> <td>c. Overall heat transfer coefficient</td> <td>d. All the options</td> </tr> </table>	a. Heat transfer coefficient	b. Thermal Conductivity	c. Overall heat transfer coefficient	d. All the options		
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5	The heat generating takes place in following equipments		1				
	<table border="1"> <tr> <td>a. Nuclear reactor</td> <td>b. Electrical conductor</td> <td>c. Chemical reactor</td> <td>d. None</td> </tr> </table>	a. Nuclear reactor	b. Electrical conductor	c. Chemical reactor	d. None		
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6	If the temperature of the solid does not change with time it is said to be at		1				

	a. Unsteady state/transient state	b. Steady State	c. Idle state	d. All the options		
7	The device which transfers energy from one fluid to the another fluid is called as					1
	a. Pump	b. Heat Exchanger	c. Mixer Grinder	d. None		
8	LMTD Stands for					1
	a. Logarithmic Mass Temperature Deviation	b. Logarithmic Mean Temperature Difference	c. Linear Mean Temperature Difference	d. None		
9	The emissivity of black body is assumed,					1
	a. Equal to unity	b. Less than unity	c. Equal to Zero	d. Equal to 10		
10	According to Kirchoff's law, at thermal equilibrium					1
	a. Absorptivity = Emissivity	b. Absorptivity < Emissivity	c. None	d. Both		
11	Following relation is true in respect of absorption (α), reflection (ρ), and transmission (τ) of radiation.					1
	a. $\alpha + \rho + \tau = 1$	b. $\alpha + \rho + \tau = 2$	c. $\alpha + \rho + \tau < 1$	d. None		
12	The ratio of actual heat transfer to the maximum possible heat transfer from a heat exchange device is called as					1
	a. Efficiency of exchanger		b. Effectiveness of exchanger			
Q. 2	Solve the following.					12
A)	Explain the various modes of heat transfer with the help of neat sketches and expressions.					6
B)	An insulating powder is densely packed in the annular space between two concentric cylinders with radii 70 mm and 45 mm, of length 1m. The inner cylinder is uniformly heated with electric power input of 1 kW. Steady state temperature attained by inner sphere is 120°C and that by outer sphere is 30°C. Neglect thermal resistance offered by the material of the spheres. (i) Calculate thermal conductivity of the powder Calculate the thermal resistance offered by insulating powder.					6

Q.3	Solve the following.		12
A)	Derive an equation for heat transfer rate through a hollow cylinder by applying Fourier's Law of Heat Conduction. The hollow cylinder has a length, L , inner radius as ' r_i ' and outer radius as ' r_o ', inner and outer wall temperatures as T_i and T_o . Thermal conductivity is represented as ' k '.		6
B)	Calculate the fin efficiency and heat loss from a fin (assuming infinitely long fin), with circular cross section under following conditions: Air temperature : 30 °C Temperature of fin at the base : 230 °C Thermal conductivity of a fin : 200 W/mK Heat transfer coefficient over a surface of a fin : 15 W/m ² K Fin Diameter : 10 mm		6
Q. 4	Solve Any Two of the following.		12
A)	Sketch the laminar and turbulent boundary layers for a flow between two parallel plates.		6
B)	Discuss the fundamentals of various types of convection with the help of real life examples.		6
C)	A solid sphere 5 cm in diameter and initially at 500°C is quenched in a controlled environment at 90°C with convective heat transfer coefficient of 115 W/m ² K. Determine the time taken by centre of the sphere to reach a temperature of 150°C. Use the properties of the solid sphere as; C_p , Sp. heat = 418 J/kgK; ρ , density = 7500 kg/m ³ , k , conductivity = 45 W/mK.		6
Q.5	Solve Any Two of the following.		12
A)	A hot rectangular plate 5 cm x 4 cm maintained at 200°C is exposed to still air at 30°C. Calculate the convective heat transfer rate by natural convection from both sides of the plate if smaller side of the plate is held vertical. Use following correlation: $Nu = 0.6 Ra^{(0.25)}$ Use the following air properties at Mean Film temperature of 115°C Density: 0.91 kg/m ³ , C_p = 1 kJ/kgK, Dynamic Viscosity (μ) 23×10^{-6} N-s/m ² , k = 0.033 W/mK.		6

B)	In a double pipe counter flow heat exchanger 165 kg/min, of an oil having a specific heat 2100 J/kgK is cooled from 90°C to 60°C by 130 kg/min of cooling water at 25°C. Assume overall heat transfer coefficient of 400 W/m ² K and specific heat of water as 4200 J/kg-K. Find out LMTD and surface area of heat exchanger.		6
C)	Discuss the terms: (i) Effectiveness of heat exchanger, (ii) NTU, (iii) Fouling factor of heat exchanger, (iv) Overall heat transfer coefficient		6
Q. 6	Solve Any Two of the following.		12
A)	Two large parallel plates, held at temperature 1000 K and 500 K, have emissivity values of 0.85 and 0.7, respectively. (i) Calculate the heat transfer rate by radiation per square meter of plate area (with no radiation shield), (ii) Calculate the heat transfer rate by radiation per square meter of plate area if the radiation shield having emissivity 0.5 is placed between the two plates Calculate the percent reduction in heat transfer rate by use of a radiation shield.		6
B)	Define any three terminologies: 1) Stefan Boltzmann Law, 2) Kirchhoff's law, 3) Radiation shape factor, 4) Laminar flow, 5) Turbulent flow, 6) Thermal Diffusivity		6
C)	Discuss the classification of Heat Exchanger with the help of neat sketches and practical examples.		6
	*** End ***		